📄 **PROJECT REPORT**

📌 **Title**

Marketing Campaign Analysis

📌 **Objective**

The objective of this project is to analyze customer data from marketing campaigns to identify high-value customers, understand factors influencing campaign response, and provide insights to improve future marketing strategies.

📌 **Tools Used**

- Python (Pandas, NumPy)
- SQL
- Power BI

📌 **Data Understanding**

The dataset contains customer-level information including demographics, spending patterns, campaign responses, and channel usage.

📌 **Data Cleaning & Transformation**

- Removed missing values
- Converted date columns
- Created new features:
    - Age
    - Total_Spend
    - Total_Purchases
    - Children

📌 **Segmentation**

Customers were segmented based on:

- Income (High / Low)
- Age (Young / Adult)
- Family (With / Without Children)

- Spending Behavior (High / Normal Spender)

📌 **Exploratory Data Analysis**

- Analyzed distribution of income, age, and spending
- Compared response rate across different segments
- Identified patterns in customer behavior

📌 **SQL Analysis**

- Created structured table for cleaned data
- Performed KPI analysis:
  - Response Rate
  - Average Spend
  - Customer Count
- Analyzed campaign performance

📌 **Dashboard**

An interactive dashboard was created using Power BI with:

- KPI cards (Total Customers, Avg Income, Response Rate)
- Charts for:
  - Income vs Response
  - Spender Type vs Response
  - Family vs Response
  - Income Distribution
- Filters:
  - Country
  - Education
  - Marital Status

📌 **Key Insights**

1. High-income customers have higher response rates
2. High spenders are more likely to respond

3. Customers without children show better engagement
4. Certain segments have high visits but low response

📌 **Recommendations**

1. Target high-income customers
2. Focus marketing on high spenders
3. Personalize campaigns based on customer segments
4. Use digital channels for better reach
5. Improve engagement strategies for low-response segments

📌 **Conclusion**

The project successfully identifies key customer segments and provides actionable insights that can help improve marketing campaign effectiveness and customer targeting strategies.